

If $\gcd(a, n) = 1$, then the unique solution to the equation $ax = 1 \pmod{n}$ is the integer x returned by EXTENDED-EUCLID, since the equation

$$\gcd(a, n) = 1 = ax + ny$$

implies $ax = 1 \pmod{n}$. Thus, EXTENDED-EUCLID can compute $a^{-1} \pmod{n}$ efficiently.

Exercises

31.4-1

Find all solutions to the equation $35x = 10 \pmod{50}$.

31.4-2

Prove that the equation $ax = ay \pmod{n}$ implies $x = y \pmod{n}$ whenever $\gcd(a, n) = 1$. Show that the condition $\gcd(a, n) = 1$ is necessary by supplying a counterexample with $\gcd(a, n) > 1$.

31.4-3

Consider the following change to line 3 of the procedure MODULAR-LINEAR-EQUATION-SOLVER:

$$3 \quad x_0 = x'(b/d) \pmod{n/d}$$

With this change, will the procedure still work? Explain why or why not.

★ 31.4-4

Let p be prime and $f(x) = (f_0 + f_1x + \cdots + f_tx^t) \pmod{p}$ be a polynomial of degree t , with coefficients f_i drawn from $\mathbb{Z}_p$. We say that $a \in \mathbb{Z}_p$ is a **zero** of f if $f(a) = 0 \pmod{p}$. Prove that if a is a zero of f , then $f(x) = (x - a)g(x) \pmod{p}$ for some polynomial $g(x)$ of degree $t - 1$. Prove by induction on t that if p is prime, then a polynomial $f(x)$ of degree t can have at most t distinct zeros modulo p .

31.5 The Chinese remainder theorem

Around 100 C.E., the Chinese mathematician Sun-Tsŭ solved the problem of finding those integers x that leave remainders 2, 3, and 2 when divided by 3, 5, and 7 respectively. One such solution is $x = 23$, and all solutions are of the form $23+105k$ for arbitrary integers k . The “Chinese remainder theorem” provides a correspondence between a system of equations modulo a set of pairwise relatively prime moduli (for example, 3, 5, and 7) and an equation modulo their product (for example, 105).

The Chinese remainder theorem has two major applications. Let the integer n be factored as $n = n_1 n_2 \cdots n_k$, where the factors n_i are pairwise relatively prime. First, the Chinese remainder theorem is a descriptive “structure theorem” that describes the structure of $\mathbb{Z}_n$ as identical to that of the Cartesian product $\mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \cdots \times \mathbb{Z}_{n_k}$ with componentwise addition and multiplication modulo n_i in the i th component. Second, this description helps in designing efficient algorithms, since working in each of the systems $\mathbb{Z}_{n_i}$ can be more efficient (in terms of bit operations) than working modulo n .

Theorem 31.27 (Chinese remainder theorem)

Let $n = n_1 n_2 \cdots n_k$, where the n_i are pairwise relatively prime. Consider the correspondence

$$a \leftrightarrow (a_1, a_2, \dots, a_k), \quad (31.27)$$

where $a \in \mathbb{Z}_n$, $a_i \in \mathbb{Z}_{n_i}$, and

$$a_i = a \bmod n_i$$

for $i = 1, 2, \dots, k$. Then, mapping (31.27) is a one-to-one mapping (bijection) between $\mathbb{Z}_n$ and the Cartesian product $\mathbb{Z}_{n_1} \times \mathbb{Z}_{n_2} \times \cdots \times \mathbb{Z}_{n_k}$. Operations performed on the elements of $\mathbb{Z}_n$ can be equivalently performed on the corresponding k -tuples by performing the operations independently in each coordinate position in the appropriate system. That is, if

$$a \leftrightarrow (a_1, a_2, \dots, a_k),$$

$$b \leftrightarrow (b_1, b_2, \dots, b_k),$$

then

$$(a + b) \bmod n \leftrightarrow ((a_1 + b_1) \bmod n_1, \dots, (a_k + b_k) \bmod n_k), \quad (31.28)$$

$$(a - b) \bmod n \leftrightarrow ((a_1 - b_1) \bmod n_1, \dots, (a_k - b_k) \bmod n_k), \quad (31.29)$$

$$(ab) \bmod n \leftrightarrow (a_1 b_1 \bmod n_1, \dots, a_k b_k \bmod n_k). \quad (31.30)$$

Proof Let's see how to translate between the two representations. Going from a to $(a_1, a_2, \dots, a_k)$ requires only k “mod” operations. The reverse—computing a from inputs $(a_1, a_2, \dots, a_k)$ —is only slightly more complicated.

We begin by defining $m_i = n/n_i$ for $i = 1, 2, \dots, k$. Thus, m_i is the product of all of the n_j 's other than n_i : $m_i = n_1 n_2 \cdots n_{i-1} n_{i+1} \cdots n_k$. We next define

$$c_i = m_i(m_i^{-1} \bmod n_i) \quad (31.31)$$

for $i = 1, 2, \dots, k$. Equation (31.31) is well defined: since m_i and n_i are relatively prime (by Theorem 31.6), Corollary 31.26 guarantees that $m_i^{-1} \bmod n_i$ exists. Here is how to compute a as a function of the a_i and c_i :

$$a = (a_1 c_1 + a_2 c_2 + \cdots + a_k c_k) \pmod{n}. \quad (31.32)$$

We now show that equation (31.32) ensures that $a = a_i \pmod{n_i}$ for $i = 1, 2, \dots, k$. If $j \neq i$, then $m_j = 0 \pmod{n_i}$, which implies that $c_j = m_j = 0 \pmod{n_i}$. Note also that $c_i = 1 \pmod{n_i}$, from equation (31.31). We thus have the appealing and useful correspondence

$$c_i \leftrightarrow (0, 0, \dots, 0, 1, 0, \dots, 0),$$

a vector that has 0s everywhere except in the i th coordinate, where it has a 1. The c_i thus form a “basis” for the representation, in a certain sense. For each i , therefore, we have

$$\begin{aligned} a &= a_i c_i && \pmod{n_i} \\ &= a_i m_i (m_i^{-1} \bmod n_i) && \pmod{n_i} \\ &= a_i && \pmod{n_i}, \end{aligned}$$

which is what we wished to show: our method of computing a from the a_i 's produces a result a that satisfies the constraints $a = a_i \pmod{n_i}$ for $i = 1, 2, \dots, k$. The correspondence is one-to-one, since we can transform in both directions. Finally, equations (31.28)–(31.30) follow directly from Exercise 31.1-7, since $x \bmod n_i = (x \bmod n) \bmod n_i$ for any x and $i = 1, 2, \dots, k$. ■

We'll use the following corollaries later in this chapter.

Corollary 31.28

If $n_1, n_2, \dots, n_k$ are pairwise relatively prime and $n = n_1 n_2 \cdots n_k$, then for any integers $a_1, a_2, \dots, a_k$, the set of simultaneous equations

$$x = a_i \pmod{n_i},$$

for $i = 1, 2, \dots, k$, has a unique solution modulo n for the unknown x . ■

Corollary 31.29

If $n_1, n_2, \dots, n_k$ are pairwise relatively prime and $n = n_1 n_2 \cdots n_k$, then for all integers x and a ,

$$x = a \pmod{n_i}$$

for $i = 1, 2, \dots, k$ if and only if

$$x = a \pmod{n}.$$

■

As an example of the application of the Chinese remainder theorem, suppose that you are given the two equations

$$a = 2 \pmod{5},$$

$$a = 3 \pmod{13},$$

so that $a_1 = 2$, $n_1 = m_2 = 5$, $a_2 = 3$, and $n_2 = m_1 = 13$, and you wish to compute $a \bmod 65$, since $n = n_1 n_2 = 65$. Because $13^{-1} = 2 \pmod{5}$ and $5^{-1} = 8 \pmod{13}$, you compute

$$c_1 = 13 \cdot (2 \bmod 5) = 26,$$

$$c_2 = 5 \cdot (8 \bmod 13) = 40,$$

and

$$\begin{aligned} a &= 2 \cdot 26 + 3 \cdot 40 \pmod{65} \\ &= 52 + 120 \pmod{65} \\ &= 42 \pmod{65}. \end{aligned}$$

	0	1	2	3	4	5	6	7	8	9	10	11	12
0	0	40	15	55	30	5	45	20	60	35	10	50	25
1	26	1	41	16	56	31	6	46	21	61	36	11	51
2	52	27	2	42	17	57	32	7	47	22	62	37	12
3	13	53	28	3	43	18	58	33	8	48	23	63	38
4	39	14	54	29	4	44	19	59	34	9	49	24	64

Figure 31.3 An illustration of the Chinese remainder theorem for $n_1 = 5$ and $n_2 = 13$. For this example, $c_1 = 26$ and $c_2 = 40$. In row i , column j is shown the value of a , modulo 65, such that $a \bmod 5 = i$ and $a \bmod 13 = j$. Note that row 0, column 0 contains a 0. Similarly, row 4, column 12 contains a 64 (equivalent to -1). Since $c_1 = 26$, moving down a row increases a by 26. Similarly, $c_2 = 40$ means that moving right by a column increases a by 40. Increasing a by 1 corresponds to moving diagonally downward and to the right, wrapping around from the bottom to the top and from the right to the left.

See Figure 31.3 for an illustration of the Chinese remainder theorem, modulo 65.

Thus, you can work modulo n by working modulo n directly or by working in the transformed representation using separate modulo n_i computations, as convenient. The computations are entirely equivalent.

Exercises

31.5-1

Find all solutions to the equations $x = 4 \pmod{5}$ and $x = 5 \pmod{11}$.

31.5-2

Find all integers x that leave remainders 1, 2, and 3 when divided by 9, 8, and 7, respectively.

31.5-3

Argue that, under the definitions of Theorem 31.27, if $\gcd(a, n) = 1$, then

$$(a^{-1} \bmod n) \leftrightarrow ((a_1^{-1} \bmod n_1), (a_2^{-1} \bmod n_2), \dots, (a_k^{-1} \bmod n_k)) .$$

31.5-4

Under the definitions of Theorem 31.27, prove that for any polynomial f , the number of roots of the equation $f(x) = 0 \pmod{n}$ equals the product of the number of roots of each of the equations $f(x) = 0 \pmod{n_1}$, $f(x) = 0 \pmod{n_2}$, ..., $f(x) = 0 \pmod{n_k}$.

31.6 Powers of an element

Along with considering the multiples of a given element a , modulo n , we often consider the sequence of powers of a , modulo n , where $a \in \mathbb{Z}_n^*$:

$$a^0, a^1, a^2, a^3, \dots,$$

modulo n . Indexing from 0, the 0th value in this sequence is $a^0 \bmod n = 1$, and the i th value is $a^i \bmod n$. For example, the powers of 3 modulo 7 are

i	0	1	2	3	4	5	6	7	8	9	10	11	...
$3^i \bmod 7$	1	3	2	6	4	5	1	3	2	6	4	5	...

and the powers of 2 modulo 7 are

i	0	1	2	3	4	5	6	7	8	9	10	11	...
$2^i \bmod 7$	1	2	4	1	2	4	1	2	4	1	2	4	...

In this section, let $\langle a \rangle$ denote the subgroup of $\mathbb{Z}_n^*$ generated by a through repeated multiplication, and let $\text{ord}_n(a)$ (the “order of a , modulo n ”) denote the order of a in $\mathbb{Z}_n^*$. For example, $\langle 2 \rangle = \{1, 2, 4\}$ in